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Real-Time Augmented Reality Item Guide

Abstract

A real-time augmented reality item guide is described. A service platform may receive a digital image depicting at least one item. A plurality of listings for the at least one item on the service platform is identified via at least one machine learning model of the service platform based on the digital image, each of the plurality of listings associated with a respective point in time. An aspect is fetched from each of the plurality of listings for the at least one item on the service platform. Time-dependent augmented reality digital content is generated for the at least one item based on the aspect. The time-dependent augmented reality digital content is rendered as an overlay on the digital image.

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Background/Summary

BACKGROUND

[0001] Search is a technique via which users of computing devices may locate information of interest. Users, for instance, may enter search queries to locate goods or services of interest or to obtain information about an item. However, conventional search techniques may involve performance of repeated queries or link selections. Moreover, search data retrieval utilizes substantial computing resources and computation power of both a client device associated with the user and a service platform executing the search, which may result in higher search latencies for a plurality of users across the service platform.

SUMMARY

[0002] A real-time augmented reality item guide is described. A service platform may receive a digital image depicting at least one item. A plurality of listings for the at least one item on the service platform is identified via at least one machine learning model of the service platform based on the digital image, each of the plurality of listings associated with a respective point in time. An aspect is fetched from each of the plurality of listings for the at least one item on the service platform. Time-dependent augmented reality digital content is generated for the at least one item based on the aspect. The time-dependent augmented reality digital content is rendered as an overlay on the digital image.

[0003] This Summary introduces a selection of concepts in a simplified form that are further described below in the Detailed Description. As such, this Summary is not intended to identify essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The detailed description is described with reference to the accompanying figures. Entities represented in the figures are indicative of one or more entities and thus reference is made interchangeably to single or plural forms of the entities in the discussion.

[0005] FIG. 1 is an illustration of a digital medium environment in an example implementation that is operable to employ augmented reality techniques described herein.

[0006] FIG. 2 depicts an example implementation showing operation of the image augmentation service of FIG. 1 in greater detail.

[0007] FIG. 3 depicts an example implementation of user interaction with an image augmentation service.

[0008] FIG. 4 depicts another example implementation of user interaction with the image augmentation service.

[0009] FIG. 5 is a flow diagram depicting an algorithm as a step-by-step procedure in an example of implementing a real-time augmented reality item guide.

[0010] FIG. 6 illustrates an example system including various components of an example device that can be implemented as any type of computing device as described and/or utilized with reference to FIGS. 1-5 to implement embodiments of the techniques described herein.

DETAILED DESCRIPTION

Overview

[0011] Advancements in computer vision, machine learning, and object recognition technologies have been used to develop augmented reality (AR) systems that digitally enhance images of a user's physical environment in real-time, such as based on images received via live camera feeds. In the

context of item searching, conventional computer vision-based image scanning techniques reduce an amount of information that is manually input by the user and enable a single item to be looked up, e.g., via a listing search of a service platform. However, the user is tasked with reviewing listing results one-by-one. As an example, a user may use a computing device (e.g., a mobile phone) to look up a value of a collectable, such as a trading card game card found at a gaming store, when making purchase decisions. However, this manual search process is time consuming and computing resource-intensive, particularly if the collectable is being offered for sale in a different language and/or currency than the user's native language and/or currency.

[0012] Accordingly, a real-time augmented reality item guide is described. This technique supports computing resource-efficient item listing search and aspect retrieval. For example, the techniques described herein enable batch-searching for a plurality of items depicted in a digital image to be performed and provide real-time information for respective items. The real-time augmented reality item guide, for instance, is generated via an image augmentation service that uses machine learning and computer vision techniques to identify an item depicted in the digital image in order to retrieve an aspect (e.g., price data, color data, size data, etc.) from a plurality of listings for the item. The image augmentation service may match the depicted item to an item identifier, such as retrieved by matching the depicted item to a cataloged item, and may further identify the plurality of listings using the item identifier. The real-time augmented reality item guide is generated based on the retrieved aspect and output as an overlay on the digital image. In the context of price, the real-time augmented reality item guide may indicate whether a price depicted in the image is more or less than the price data for the plurality of listings. By retrieving aspects from the listings rather than loading each listing and its associated data individually, a burden on computing resources is reduced.

[0013] In this way, a number of repeated inputs used to obtain item aspect information is reduced, which decreases network traffic and makes computing resources available for performing other tasks, resulting in lower response latencies across a service platform. Additionally, by processing the reduced number of inputs, less power is used, thereby increasing device battery life and/or enabling the power to be used for alternative tasks. In this way, the real-time augmented reality item guide described herein improves operation of a computing device. Further discussion of these and other examples is included in the following discussion and shown in corresponding figures.

[0014] In some aspects, the techniques described herein relate to a method for an augmented reality guide, the method including: receiving, by a service platform, a digital image depicting at least one item; identifying, via at least one machine learning model of the service platform, a plurality of listings for the at least one item on the service platform based on the digital image, each of the plurality of listings associated with a respective point in time; fetching an aspect from each of the plurality of listings for the at least one item on the service platform; generating time-dependent augmented reality digital content for the at least one item based on the aspect; and rendering the time-dependent augmented reality digital content as an overlay on the digital image.

[0015] In some aspects, the techniques described herein relate to a method, wherein the aspect includes at least one of a color, a size, a brand, a model, or a year of release of the at least one item.

[0016] In some aspects, the techniques described herein relate to a method, wherein the aspect includes price data, and the method further includes: detecting, via a classifier, a price of the at least one item as depicted in the digital image; and generating the time-dependent augmented reality digital content for the at least one item further based on the detected price.

[0017] In some aspects, the techniques described herein relate to a method, wherein the time-dependent augmented reality digital content includes a first visual indicator in response to the detected price being less than an average of the price data or a second visual indicator in response to the detected price being greater than the average of the price data.

[0018] In some aspects, the techniques described herein relate to a method, wherein the time-dependent augmented reality digital content includes an indication of the price of the at least one

item in a different currency than the detected price.

[0019] In some aspects, the techniques described herein relate to a method, wherein the time-dependent augmented reality digital content includes a summary of the price data.

[0020] In some aspects, the techniques described herein relate to a method, wherein the summary includes an average price, a highest price, and a lowest price of the plurality of listings.

[0021] In some aspects, the techniques described herein relate to a method, wherein identifying, via the at least one machine learning model of the service platform, the plurality of listings for the at least one item on the service platform based on the digital image includes: detecting a boundary of the at least one item based on the digital image; outputting feature vectors of a portion of the digital image within the detected boundary; and matching the feature vectors to an item listed on the service platform via the plurality of listings.

[0022] In some aspects, the techniques described herein relate to a method, wherein the at least one machine learning model of the service platform is trained for a category of items that includes the at least one item, and wherein the category of items is received via a user interface.

[0023] In some aspects, the techniques described herein relate to a method, wherein the digital image is received as part of a live camera feed.

[0024] In some aspects, the techniques described herein relate to a method, further including: receiving a selection of the time-dependent augmented reality digital content; and generating a chart for the aspect with respect to a defined period of time based on aspect data for listings associated with time points within the defined period of time, the defined period of time adjustable based on an input.

[0025] In some aspects, the techniques described herein relate to a method, wherein the at least one item includes a plurality of items, and wherein the time-dependent augmented reality digital content is generated for individual items of the plurality of items in parallel.

[0026] In some aspects, the techniques described herein relate to a computing device including: a processing system; and a computer-readable storage medium storing instructions that, responsive to execution by the processing system, causes the processing system to perform operations including: displaying a digital image from a live camera feed; determining an item identifier for at least one object depicted in the digital image by processing the digital image using a computer vision module of a service platform; retrieving, based on the item identifier, aspect data from a plurality of listings for the at least one object on the service platform, each of the plurality of listings associated with a respective time point; generating augmented reality digital content for the at least one object based on the aspect data with respect to a specified period of time; and rendering the augmented reality digital content as an overlay on the digital image.

[0027] In some aspects, the techniques described herein relate to a computing device, wherein determining the item identifier for the at least one object depicted in the digital image by processing the digital image using the computer vision module of the service platform includes: detecting a boundary of the at least one object in the digital image; outputting feature vectors of a portion of the digital image within the detected boundary; and matching the feature vectors to an item having the item identifier via a similarity measurement, wherein the plurality of listings includes instances of the item having the item identifier listed for sale on the service platform.

[0028] In some aspects, the techniques described herein relate to a computing device, wherein generating the augmented reality digital content for the at least one object based on the aspect data with respect to the specified period of time includes: receiving, by the service platform, a selection of the specified period of time; constraining the aspect data to a portion generated within the specified period of time based on the respective time point of an associated listing of the plurality of listings; and generating the augmented reality digital content using the constrained aspect data.

[0029] In some aspects, the techniques described herein relate to a computing device, wherein the augmented reality digital content indicates at least one of an average sale price, a highest sale price, a lowest sale price, a purchase profitability, a number sold, a color, a size, a brand, a model, or a

year of release of the at least one object within the specified period of time.

[0030] In some aspects, the techniques described herein relate to a computer-readable storage medium storing instructions that, responsive to execution by one or more processors, causes the one or more processors to perform operations including: receiving, by a service platform, a digital image from a live camera feed; determining an item identifier for an item depicted in the digital image by processing the digital image via at least one machine learning model of the service platform; retrieving a plurality of listings for the item on the service platform based on the item identifier; fetching aspect data from each of the plurality of listings for the item on the service platform, the aspect data associated with points in time corresponding to respective listings of the plurality of listings; generating time-dependent augmented reality digital content for the item based on the aspect data; and rendering the time-dependent augmented reality digital content as an overlay on the digital image.

[0031] In some aspects, the techniques described herein relate to a computer-readable storage medium, wherein determining the item identifier for the item depicted in the digital image by processing the digital image via the at least one machine learning model of the service platform includes: detecting a boundary of the item in the digital image via an object detection algorithm; extracting feature vectors from within the detected boundary via a feature extraction algorithm; and matching the feature vectors to an item having the item identifier via an item identification algorithm and based on a catalog of items, the catalog of items including entries for a plurality of items associated with item identifiers.

[0032] In some aspects, the techniques described herein relate to a computer-readable storage medium, wherein the operations further include: selecting the at least one machine learning model of the service platform based at least in part on a category input.

[0033] In some aspects, the techniques described herein relate to a computer-readable storage medium, wherein generating the time-dependent augmented reality digital content for the item based on the aspect data includes: generating a chart relating an aspect of the aspect data to time for a defined period of time, the defined period of time adjustable based on an input.

[0034] In the following discussion, an example environment is first described that may employ the techniques described herein. Examples of implementation details and procedures are then described which may be performed in the example environment as well as other environments. Performance of the example procedures is not limited to the example environment, and the example environment is not limited to performance of the example procedures.

Example of an Environment

[0035] FIG. 1 is an illustration of a digital medium environment **100** in an example implementation that is operable to employ real-time augmented reality item guide techniques described herein. The illustrated environment **100** includes a service provider system **102** and a client device **104** of a user **106** that are communicatively coupled, one to another, via a network **108**. Computing devices that implement the service provider system **102** and the client device **104** may be configured in a variety of ways.

[0036] A computing device, for instance, is configurable as a desktop computer, a laptop computer, a mobile device (e.g., assuming a handheld configuration such as a tablet or mobile phone), and so forth. Thus, a computing device ranges from full resource devices with substantial memory and processor resources (e.g., personal computers, game consoles) to a low-resource device with limited memory and/or processing resources, e.g., mobile devices. Additionally, a computing device is also representative of a plurality of different devices, such as multiple servers utilized by a business to perform operations “over the cloud” as illustrated for the service provider system **102** and as described with respect to FIG. 6.

[0037] The service provider system **102** includes an executable service platform **110**. The executable service platform **110** is configured to implement and manage access to digital services **112** “in the cloud” that are accessible by the user **106** via the client device **104** and the network **108**.

Thus, the executable service platform **110** provides an underlying infrastructure to manage execution of the digital services **112**, e.g., through control of underlying computational resources. [0038] The executable service platform **110** supports numerous computational and technical advantages, including an ability of the service provider system **102** to readily scale resources to address wants of the user **106** as well as other entities. Thus, instead of incurring an expense of purchasing and maintaining proprietary computer equipment for performing specific computational tasks, cloud computing provides the user **106** with access to a wide range of hardware and software resources so long as the client device **104** has access to the network **108**.

[0039] The digital services **112** can take a variety of forms. Examples of digital services include social media services, document management services, storage services, media streaming services, content creation services, productivity services, digital marketplace services, auction services, and so forth. In the present example, the digital services **112** include an image augmentation service **114**. In at least one implementation, the image augmentation service **114** includes functionality for implementing real-time augmented reality (AR) guides for at least one item depicted in a digital image. As used herein, “real-time” or “live” refers to processes that occur substantially instantaneously, without intentional delay. The image augmentation service **114** performs image scanning tasks in order to detect and identify item(s) depicted in a digital image, such as received from a live camera feed from the client device **104**, using a machine learning platform **116**. As will be elaborated herein, upon recognition of the item(s) via the machine learning platform **116**, the image augmentation service **114** retrieves item details using a search module **118** and generates AR content via a digital content generation module **120** based on data retrieved via the search module **118**.

[0040] The client device **104**, for instance, may include an input device **122** that is usable to generate input data **124**, such as text via a keyboard, a spoken utterance that is converted to text, a capture of one or more digital images **126** via a camera platform **128**, and so forth. The input data **124** is then communicated by a communication module **130** (e.g., a browser or a network-enabled application) to the image augmentation service **114** via the network **108**. For example, the user **106** may access and/or interact with the camera platform **128** via the input device **122**. The camera platform **128**, for instance, may implement a “live view” formed of digital images **126** taken of a physical environment of the client device **104** (e.g., a real-time camera feed) and provide this live view as at least a portion of the input data **124** to the image augmentation service **114**. Additional examples of the input data **124** are described herein, e.g., with respect to FIG. 2.

[0041] The machine learning platform **116** may utilize one or more machine learning models, including different types of machine learning models where the underlying models are learned using different approaches, such as using supervised learning, unsupervised learning, and/or reinforcement learning. By way of example, these models may include regression models (e.g., linear, polynomial, and/or logistic regression models), classifiers, Bayesian networks, random forest learning, dimensionality reduction algorithms, boosting algorithms, neural networks, deep learning models, and reinforcement learning-based models, to name just a few. To process the digital images **126**, the machine learning platform **116** of the image augmentation service **114** includes a computer vision module **132** that includes functionality to perform object detection, object recognition, and classification in order to gain a high-level understanding of the visual information included in the digital images **126**. Models and/or algorithms of the computer vision module **132** may be trained via one or a combination of supervised learning (where target output labels are included in the training data), unsupervised learning (where explicit target output labels are not provided in the training data), self-supervised learning (where the computer vision module **132** generates the target output labels from the training data), and semi-supervised learning (where the training data includes a combination of labeled and unlabeled examples).

[0042] Thus, the machine learning platform **116** may employ at least one model and/or algorithm that is tuned (e.g., trained) based on inputs to approximate unknown functions and provide a

desired output. This may include employing computer functionality to learn from, and make predictions on, known data by analyzing the known data to learn to generate outputs that reflect patterns and attributes of the known data. Once trained, the at least one model and/or algorithm is usable to “understand” novel data, e.g., the digital images **126**, by generating data-driven predictions or decisions based on the input data **124**.

[0043] In the example illustrated in FIG. **1**, the computer vision module **132** includes an object detection algorithm **134**, a feature extraction algorithm **136**, and an item identification algorithm **138**. The object detection algorithm **134**, for instance, is trained to locate and identify (e.g., classify) one or more objects within the digital images **126**, such as based on their shape. For example, the object detection algorithm **134** may be configured to detect a boundary of an object depicted in the digital images **126**. The feature extraction algorithm **136** may be trained to identify specific patterns, edges, colors, textures, textual features, or other distinctive visual elements from the one or more objects identified by the object detection algorithm **134**. In this context, “extracting” refers to identifying relevant information from the digital images **126** in order to isolate relevant attributes, e.g., “features,” which are typically represented as feature vectors that are numerical representations of the input data. For example, the feature extraction algorithm **136** may be configured to output feature vectors for a portion of the digital images **126** within the boundary detected by object detection algorithm **134**.

[0044] Once the detected object(s) in the digital images **126** are represented as feature vectors, the item identification algorithm **138** uses the feature vectors as input for determining an item identity of the detected object(s). By way of example, the item identification algorithm **138** matches an object detected in the digital images **126** to a known, cataloged item by referencing a catalog **140**. In at least one implementation, the catalog **140** is a comprehensive database that provides detailed information about a collection of items, such as coins, stamps, trading cards (e.g., sports cards or a cards of a trading card game), limited edition items, designer items (e.g., watches, purses, etc.), comics, toys, figurines, apparel, or the like. Entries for individual items in the collection may be stored on the executable service platform **110** in the catalog **140** according to an item identifier (ID), which includes a unique code or string of characters (e.g., alphanumeric characters) that enables the associated entry to be identified and referenced. The entries of the catalog **140** are depicted in FIG. **1** as a first item **142(1)** (e.g., “item **1**”), a second item **142(2)** (e.g., “item **2**”), and an nth item **142(N)** (e.g., “item **N**”), with ellipses denoting that one or more other items may exist in the catalog **140**.

[0045] The catalog **140** is shown as being stored in a storage device **144**, e.g., a data warehouse of the service provider system **102**. Although one catalog **140** is shown, it is to be appreciated that the storage device **144** may store a plurality of catalogs. For instance, respective catalogs of the plurality of catalogs are associated with different collections. By way of example, a first catalog systematically provides information for a first collection of items (e.g., a particular trading card game), and a second catalog systematically provides information for a second collection of items (e.g., watches by a specific designer).

[0046] The item identification algorithm **138** may reference the catalog **140** to perform feature vector matching between the feature vectors generated by the feature extraction algorithm **136** and feature vectors of items in the catalog. A “match” in this context thus refers to the feature vectors of the detected object, as output by the feature extraction algorithm **136**, being the same as feature vectors for an item in the catalog, such as via a similarity measurement technique. In at least one implementation, the computer vision module **132** includes a plurality of different item identification algorithms that are trained to match items in different catalogs or collections of catalogs stored in the storage device **144**. For instance, a first item identification algorithm may be trained to perform feature vector matching for trading cards, a second item identification algorithm may be trained to perform feature vector matching for watches, and so forth. By performing the feature vector match, the item identification algorithm **138** may identify the item ID of the object

detected in the digital images **126**.

[0047] The machine learning platform **116** may further include a text processing module **146** that utilizes natural language processing (NLP) techniques and machine learning models, such as recurrent neural networks (RNNs), transformers, and optical character recognition to detect and understand text (e.g., alphanumerical characters) depicted in the digital images **126**. By way of example, the text processing module **146** includes a large language model (LLM) that is initially trained (e.g., pre-trained) on a large corpus of diverse textual data to learn patterns and relationships within the textual data, allowing the text processing module **146** to perform tasks like translation, summarization, and so forth. In at least one implementation, the LLM is further trained (e.g., fine-tuned) using the catalog **140** so that the text processing module **146** is specialized for data that is specific to items in the catalog **140**. During the fine-tuning, for instance, weights and parameters of the LLM are adjusted using training data generated from the catalog **140** using one or a combination of supervised learning (where target output labels are included in the training data), unsupervised learning (where explicit target output labels are not provided in the training data), self-supervised learning (where the machine learning platform **116** generates the target output labels from the training data), and semi-supervised learning (where the training data includes a combination of labeled and unlabeled examples). It is to be appreciated that the text processing module **146** may include a plurality of models that perform different specialized tasks (e.g., a first model for language detection, a second model for language translation, a third model for currency conversion, a fourth model for context understanding, etc.) rather than a singular model that performs an entirety of the language processing tasks.

[0048] As a non-limiting, illustrative example, the object detection algorithm **134** detects a price tag associated with an object in the digital images **126**, e.g., via a price tag classifier. The text processing module **146** detects numerical values included on the detected price tag and further determines a currency of a price shown on the price tag. The text processing module **146** converts the price to a desired currency based on receiving the desired currency as part of the input data **124**. The converted price may then be used by the digital content generation module **120** to generate AR digital content, as will be elaborated below.

[0049] Once the item identification algorithm **138** identifies the item ID of the object detected in the digital images **126** from the catalog **140**, the search module **118** identifies corresponding listings in a listing database **148**, which is illustrated as being stored in a storage device **150** (e.g., a data warehouse of the executable service platform **110**) using the item ID. In one or more implementations, individual listings for the plurality of listings are stored as entries in the listing database **148** according to a listing identifier (ID), which includes a unique code or string of characters (e.g., alphanumeric characters) that enables the associated listing to be tracked, organized, and managed. The entries of the listing database **148** are depicted in FIG. **1** as a first listing **152(1)** (e.g., “listing 1”), a second listing **152(2)** (e.g., “listing 2”), and an n.sup.th listing **152(N)** (e.g., “listing N”), with ellipses denoting that one or more other listings may exist in the listing database **148**. It is to be appreciated that individual entries in the listing database **148** include different listing IDs.

[0050] In at least one implementation, the search module **118** performs batch-searching of a plurality of items identified in the digital images **126**, which reduces computational resources and network traffic compared to searching for the items individually as well as reducing an amount of time performed searching. By way of example, if ten items are identified in the digital images **126**, a single search is performed via the search module **118** rather than ten individual queries. In this way, an operating efficiency of the service provider system **102** is increased. Furthermore, power consumption by the client device **104** is decreased by reducing repeated user inputs, which thus improves the operation of the client device **104**.

[0051] The entries in the listing database **148** include instances of items that have been registered on the executable service platform **110** (e.g., for sale) in order to make the items available to other

users, e.g., for purchase. The listing database **148** includes, for example, active listings of items currently available and purchasable by the user **106** as well as inactive listings of items that are no longer available and purchasable by the user **106**, such as due to a sale of the item via the listing or an expiration of the listing. It is to be appreciated that multiple listings may exist in the listing database **148** for a specific item, such as when multiple instances of the item are listed for sale on the service provider system **102**. In contrast to this, the catalog **140** includes one entry for the specific item.

[0052] In accordance with the techniques described herein, in response to receiving the item ID for at least one item depicted in the digital images **126** from the item identification algorithm **138**, the search module **118** identifies (e.g., selects) listings of the listing database **148** for the at least one item and fetches aspect data **154** describing one or more aspects from each of the identified listings via an aspect retrieval function **156**. Non-limiting examples of aspects include price, size, color, model, brand, designer, year of manufacture or release, issue number, and so forth. The aspect retrieval function **156** includes functionality for extracting aspect data (e.g., as a datafile or list) from the identified listings, which results in a smaller amount of data being transferred (e.g., over the network **108**) compared with individually loading the identified listings, such as occurs during a manual search performed by the user **106**. Moreover, in at least one implementation, the aspect retrieval function **156** performs batch retrieval for the aspect data **154** for of a plurality of items identified in the digital images **126**.

[0053] In one or more implementations, the listings in the listing database **148** are each associated with a respective point in time, such as a time at which the corresponding listing was generated or the associated item was sold. As such, the aspect data **154** retrieved from the listing database **148** may be associated with the respective points in time in order to generate time-dependent data. The time-dependence, for example, may account for how an aspect, such as the color or price, of the associated item varies or remains constant (e.g., does not vary) with respect to time. This time-dependence may be used by the digital content generation module **120** based on preferences received via the input data **124**, for example.

[0054] The aspect data **154** are used by the digital content generation module **120** to generate augmented reality digital content **158** (abbreviated as “AR digital content” in the figures), which is communicated to the client device **104**, e.g., via the communication module **130**. The augmented reality digital content **158**, for instance, may describe aspects of the at least one item, such as a brand name of the at least one item, a price for which the at least one item is available for sale or purchase (e.g., via the service provider system **102**), a relative popularity of the at least one item (e.g., based on a number sold during a specified period of time), and so forth, as determined from the aspect data **154**. Accordingly, the augmented reality digital content **158** may function as an item guide **160** that provides information regarding the at least one item and/or for purchasing the at least one item. The item guide **160**, for instance, may include a price guide that shows how a monetary value of the object has changed with respect to time, as will be further elaborated herein, e.g., with respect to FIG. 4.

[0055] To generate the augmented reality digital content **158**, the digital content generation module **120** may include functionality for interpreting and/or summarizing the aspect data **154** retrieved from the listing database **148**. By way of example, the digital content generation module **120** may be configured to generate an average sold price of the item within a defined time period (e.g., within the last month or the last year), output a number of items sold within the defined time period, and so forth as a summary of the aspect data **154**. As such, the augmented reality digital content **158** may be time-dependent and vary as a function of time, e.g., as new data becomes available or data used in generating the digital content becomes out of date. In this context, the augmented reality digital content **158** may include real-time, up-to-date information.

[0056] The augmented reality digital content **158** is rendered for display at the client device **104**, such as via a display device **162** of the client device **104**. In at least one implementation, the

augmented reality digital content **158** is rendered as an overlay on the digital images **126**. For example, the augmented reality digital content **158** is displayed proximal to the at least one item. In this way, the camera platform **128** supports functionality for the user **106** to “look around” the physical environment and display aspect information that describes objects included within the physical environment. Moreover, the augmented reality digital content **158** may include user-selectable elements that enable different or additional information to be displayed.

[0057] In general, functionality, features, and concepts described in relation to the examples above and below are employed in the context of the example procedures described in this section. Further, functionality, features, and concepts described in relation to different figures and examples in this document are interchangeable among one another and are not limited to implementation in the context of a particular figure or procedure. Moreover, blocks associated with different representative procedures and corresponding figures herein are applicable together and/or combinable in different ways. Thus, individual functionality, features, and concepts described in relation to different example environments, devices, components, figures, and procedures herein are usable in any suitable combinations and are not limited to the particular combinations represented by the enumerated examples in this description.

Real-Time Augmented Reality Item Guide

[0058] FIG. 2 depicts an example implementation **200** showing operation of the image augmentation service **114** of FIG. 1 in greater detail. The following discussion describes techniques that are implementable utilizing the previously described systems and devices, and thus, where applicable, reference will be made to components previously introduced in FIG. 1.

[0059] In the example implementation **200**, the client device **104** receives profile data **202** via a profile data collection module **204** to generate a user profile **206**. The user profile **206**, for instance, includes preferences for customizing information included in the augmented reality digital content **158** and/or the way the information is displayed. For instance, the user profile **206** may include input from the user **106** regarding visualization preferences (e.g., color schemes, text sizes, etc.) as well as aspects to include in the augmented reality digital content **158**. Additionally or alternatively, the user profile **206** may include collection data, where the user **106** specifies which items in the catalog **140** are already owned, and/or wish list data, where the user **106** specifies which items in the catalog **140** are desired. Although the user profile **206** is shown as being stored on the client device **104**, it is to be appreciated that in other implementations, the user profile **206** is additionally or alternatively stored on the service provider system **102**. The profile data collection module **204** may receive the profile data **202** via the input device **122**, for example.

[0060] The user profile **206** is included in the input data **124** communicated to the image augmentation service **114**. As described above with respect to FIG. 1, the input data **124** additionally include the digital images **126**, such as obtained via the camera platform **128**. In the example implementation **200**, the input data **124** further include a category **208**, a desired currency **210**, and a desired time period **212**. The category **208** describes a type of object (or item) that is to be identified via the image augmentation service **114**. The desired currency **210** indicates a selected currency in which the user **106** wishes to receive price information, when included, and the desired time period **212** indicates a date and/or time range for constraining the search module **118**. That is, search module **118** may retrieve listings associated with timepoints within the desired time period **212** so that aspects from listings associated with timepoints outside of the desired time period **212** are not used in generating the augmented reality digital content **158**. It is to be appreciated that additional or alternative inputs may be included in the input data **124** without departing from the scope of the described techniques. Moreover, in at least one implementation, one or more or each of the category **208**, the desired currency **210**, and the desired time period **212** is included in the user profile **206**.

[0061] The user **106** may use the input device **122** of the client device **104** to select or otherwise input the category **208**, the desired currency **210**, and the desired time period **212**. In some

instances, the category **208**, the desired currency **210**, and the desired time period **212** may be automatically populated based on a default setting or stored preferences of the user **106** in the user profile **206**. The user **106** may further adjust the automatically populated information via the input device **122**, as desired.

[0062] As an illustrative example, the user **106** may use the input device **122** to select or otherwise input “trading cards” as the category **208** when capturing digital images of trading cards, “USD” as the desired currency **210** when the user **106** wants to view monetary values in United States dollars, and “past month” as the desired time period **212** when the user **106** is interested in trading card sales that have occurred in the previous month. Based on these inputs, the image augmentation service **114** may be configured to identify trading cards in the digital images **126** and generate augmented reality digital content **158** for the trading cards based on listing data associated with points in time within the previous month, with monetary information shown in the augmented reality digital content **158** provided in USD. Accordingly, the category **208**, the desired currency **210**, and the desired time period **212** may affect how the digital images **126** are processed via the computer vision module **132**, how the search is performed by the search module **118**, and/or how the augmented reality digital content **158** is generated by the digital content generation module **120**.

[0063] The image augmentation service **114** receives the input data **124**, e.g., from the client device **104** via the communication module **130**. In at least one implementation, the computer vision module **132** includes different object detection algorithms **134** and/or different item identification algorithms **138** for different categories, and so the category **208** may instruct the computer vision module **132** regarding which algorithm(s) or catalog(s) to utilize. Additionally or alternatively, the category **208** may instruct the computer vision module **132** to identify objects in the digital images **126** that are included in the given category while ignoring others. As an illustrative example where the category **208** is “watches” and the digital images **126** depict watches among other accessories (e.g., necklaces and earrings), the computer vision module **132** may identify item IDs for the watches and not the other accessories.

[0064] The computer vision module **132** performs object detection via the object detection algorithm **134** based on, e.g., the digital images **126** and the category **208**. By way of example, the object detection algorithm **134** may use computational logic to detect shapes of objects in the digital images **126** that are consistent with known shapes of objects in the category **208**. As a non-limiting example where the category **208** is trading cards, the object detection algorithm **134** may detect trading cards in the digital images **126** based on their rectangular shapes. The object detection algorithm **134** outputs at least one detected object **214**, including its detected boundary.

[0065] The detected object **214** is further evaluated by the feature extraction algorithm **136** to generate feature vectors **216**. By way of example, the feature extraction algorithm **136** evaluates pixels within the detected boundary of the detected object **214** to identify specific patterns, edges, colors, textures, textual features, or other distinctive visual elements, which are extracted as the feature vectors **216**.

[0066] The item identification algorithm **138** matches the feature vectors **216** to an item in the catalog **140**. By way of example, the item identification algorithm **138** may compare the feature vectors **216** with feature vectors for items in the catalog **140** using a similarity measurement. The similarity measurement may use a Euclidean distance, a cosine similarity measurement, and/or a correlation coefficient, for example. For instance, a smaller Euclidean distance may indicate a higher similarity value, and thus a closer match between the feature vectors **216** of the detected object **214** and the item in the catalog **140**. Additionally or alternatively, a similarity threshold may be used. By way of example, the similarity measurements for feature vector comparisons between the feature vectors **216** and those for an item in the catalog **140** may be averaged (e.g., as a weighted or non-weighted average), and a match may be identified when the average similarity measurement is greater than a pre-determined similarity threshold. When a match is identified, e.g., when the average similarity measurement is greater than the similarity threshold, an item ID **218**

associated with the matching item is retrieved from the catalog **140**.

[0067] In the example implementation **200**, the object detection algorithm **134** further outputs a detected price tag **220**. By way of example, the object detection algorithm **134** includes a price tag classifier that is trained to identify price tags on detected objects, such as based on their size, shape, and the content included, such as numerical price. The detected price tag **220** is received by the text processing module **146**, which outputs a price **222** in the desired currency **210**. In some scenarios, the price **222** is in a different currency than in the detected price tag **220**. As such, in at least one implementation, the text processing module **146** identifies a depicted currency in the detected price tag **220** and performs a currency conversion to the desired currency **210** according to a current exchange rate between the depicted currency and the desired currency **210**.

[0068] The item ID **218** is received by the search module **118**, which identifies listings **224** of the listing database **148** for items having the item ID **218** within the desired time period **212**. The aspect retrieval function **156** further fetches the aspect data **154** from the listings **224**. The aspect data **154** includes information regarding at least one aspect of the item having the item ID **218**, as provided in the listings **224**. For instance, the at least one aspect may be a size, a color, a sale price, a brand, a model, or a year of release, such as according to a default for the category **208** and/or based on the user profile **206**. Because each listing in the listing database **148** is associated with a respective point in time, the aspect data **154** associates the at least one aspect with the respective point in time.

[0069] The digital content generation module **120** receives the aspect data **154** and the price **222**, when detected, and generates the augmented reality digital content **158** therefrom. The augmented reality digital content **158**, for instance, may describe characteristics of the detected and identified object, a brand name of the detected and identified object, a price for which the detected and identified object is available for sale or purchase (e.g., via an online auction), and so forth, based on the particular aspect(s) included aspect data **154**. The augmented reality digital content **158** may additionally include recommendations, digital marketing content configured to cause conversion of a product or service, streaming digital media, digital music, search results, links, and so forth. The augmented reality digital content **158** is configured to be displayed proximal to the detected and identified object as an overlay on the digital images **126**, for example. In at least one implementation, the augmented reality digital content **158** includes selectable options that are configured to output additional digital content, such as will be elaborated herein with respect to FIG. 4.

[0070] In at least one implementation, the digital content generation module **120** additionally uses the input data **124** to customize information included in the augmented reality digital content **158** and/or the way the information is displayed. For instance, the augmented reality digital content **158** may include price information, including the price **222** of the detected price tag **220** and/or a sale price generated from the aspect data **154**, in the desired currency **210**. Additionally or alternatively, the input data **124** may be used to adjust colors, sounds, and graphics used in the augmented reality digital content **158**. Non-limiting examples of how the augmented reality digital content **158** is visually rendered are described below with respect to FIG. 4.

[0071] Referring now to FIG. 3, an example implementation **300** of user interaction with the image augmentation service **114** is shown. The example implementation **300** is illustrated using a first stage **302**, a second stage **304**, and a third stage **306**.

[0072] The example implementation **300** depicts the client device **104** displaying a user interface **308**. In one or more implementations, the user interface **308** is an interface included in a series or sequence of user interfaces output while accessing the image augmentation service **114** via the client device **104**, e.g., via the communication module **130** depicted in FIG. 1.

[0073] At the first stage **302**, the user interface **308** is output by the client device **104**, e.g., via the display device **162**, which may be a touchscreen display device. By way of example, the touchscreen display device may be configured to both display outputs of the image augmentation

service **114** and receive inputs to be used by the image augmentation service **114**. As such, the touchscreen display device may be configured as both the display device **162** and the input device **122**. The user interface **308** is configured to display a “live feed” of digital images **126** obtained in real-time from the camera platform **128** in this example.

[0074] The live feed of the digital images **126** depicts a plurality of objects **310** in a physical environment. In the illustrated example, the plurality of objects **310** are trading cards. The user interface **308** also includes a selection menu **312** that includes prompts for receiving input to the image augmentation service **114**. Input is received, for example, via a hand **314** (e.g., a finger of the user **106**) interacting with the touchscreen functionality of the client device **104**. In variations, the input is additionally or alternatively received via text commands, spoken commands, mouse or joystick movements, keystrokes, buttons, or the like.

[0075] At the first stage **302** of the illustrated example, the selection menu **312** includes a currency selection option **316** and a category selection option **318**. The currency selection option **316** enables the user **106** to select a desired output currency (e.g., the desired currency **210**), while the category selection option **318** enables the user **106** to select a category of the plurality of objects **310**, e.g., the category **208**. As described above with respect to FIG. 2, the desired currency **210** indicates a currency for displaying monetary or price information, and the category **208** describes a type of object (or item) that is to be identified via the image augmentation service **114**. It is to be appreciated that the implementation **300** shows one example of the options that may be included in the selection menu **312**, and the selection menu **312** may include additional or different options without departing from the scope of the described techniques.

[0076] The selection menu **312** further includes a control element **320**. For example, selection of the control element **320** may cause the image augmentation service **114** to perform one or more tasks, such as object identification via the computer vision module **132**. In the first stage **302**, the control element **320** is diagonally shaded to indicate that the control element **320** is not selectable at the first stage **302**. For example, the control element **320** may not be selectable until input is received from the user **106** via the currency selection option **316** and the category selection option **318**. Accordingly, the first stage **302** shows the hand **314** interacting with the category selection option **318** via a finger tap. As such, user input is received at the category selection option **318** that selects the category selection option **318** for additional user interaction.

[0077] In response to receiving selection of the category selection option **318** at the first stage **302**, a category selection menu **322** is opened at the second stage **304**. The category selection menu **322** provides selectable options for the category **208**, e.g., “comic books,” “trading cards,” and “purses” in the example shown. In variations, the selectable options include additional or alternative categories to those shown. Moreover, in at least one variation, the selectable options include sub-categories, such as a type of trading card, e.g., a sport or particular trading card game.

[0078] At the second stage **304**, user input is detected at “trading cards,” e.g., via a finger of the hand **314** tapping “trading cards.” Selection of “trading cards” is indicated by a check in the present example. Thus, via the category selection menu **322**, the image augmentation service **114** receives instructions to generate the augmented reality digital content **158** with respect to trading cards identified in the plurality of objects **310**.

[0079] The third stage **306** shows a currency selection menu **324**. By way of example, the currency selection menu **324** is opened in response to receiving user input at the currency selection option **316**. The currency selection menu **324** provides selectable options for the desired currency **210**, e.g., “JP yen,” “UK pound sterling,” and “US dollar” in the example shown. In variations, the selectable options include additional or alternative currencies to those shown. At the third stage **306**, user input has been received to select “US dollar,” as indicated by a check next to corresponding option. Moreover, the category selection option **318** shows that trading cards remains selected. As such, input has been received at both the currency selection option **316** and the category selection option **318**. Accordingly, the control element **320** is selectable at the third stage

306, as indicated by the absence of diagonal shading.

[0080] Upon receiving user input at the control element **320**, for instance, the image augmentation service **114** may process the digital images **126** via the computer vision module **132** as described above to identify one or more trading cards in the plurality of objects **310**. In this example, the processing includes the object detection algorithm **134** detecting trading cards based on their rectangular shape, and the feature extraction algorithm **136** outputs feature vectors **216** for individual trading cards. The item identification algorithm **138** then uses these feature vectors **216** to retrieve the item ID **218** for one or more detected trading cards that have matching entries in the catalog **140**. Based on the identified one or more trading cards, the image augmentation service **114** fetches aspect data from the listing database **148** via the search module **118** and the aspect retrieval function **156** and generates augmented reality digital content **158** via the digital content generation module **120**, as discussed above and elaborated below with respect to FIG. 4.

[0081] FIG. 4 depicts another example implementation **400** of user interaction with the image augmentation service **114**. While FIG. 3 is directed to receiving user inputs for generating the augmented reality digital content **158** by the image augmentation service **114**, FIG. 4 depicts user interaction with the augmented reality digital content **158**. The example implementation **400** is illustrated using a first stage **402**, a second stage **404**, and a third stage **406**.

[0082] The example implementation **400** depicts the client device **104** displaying the user interface **308** introduced above with respect to FIG. 3 displaying the “live feed” of the digital images **126** of the plurality of objects **310**. At the first stage **402**, the user interface **308** also displays the augmented reality digital content **158**. For example, the augmented reality digital content **158** is rendered as an overlay on the digital images **126**.

[0083] In the illustrated example, the augmented reality digital content **158** includes a plurality of different visual indicators. As a non-limiting example where the augmented reality digital content **158** is related to price (e.g., price is the aspect), the augmented reality digital content **158** includes a first visual indicator **408** for detected and identified trading cards (e.g., individual trading cards that have been matched to an item ID in the catalog **140** by the item identification algorithm **138**) where a detected price is less than an average price retrieved from the listing database **148**. The first visual indicator **408** is illustrated as a solid bold line around the individual trading card, although other types of visual indications are possible. By way of example, the first visual indicator **408** may additionally or alternatively include a first color, such as green. For illustrative clarity, the first visual indicator **408** is labeled for one trading card in the first stage **402**. The augmented reality digital content **158** further includes a second visual indicator **410** for detected and identified trading cards where the detected price is greater than the average price retrieved from the listing database **148**. The second visual indicator **410** is illustrated as a bold dashed lined around the individual trading card, although other types of visual indications are possible. By way of example, the second visual indicator **410** may additionally or alternatively include a second color, such as red. For illustrative clarity, the second visual indicator **410** is labeled for one trading card in the first stage **402**.

[0084] Via the first visual indicator **408** and the second visual indicator **410**, information regarding a potential profit or loss of purchasing the detected and identified trading card is quickly and simply conveyed, e.g., to the user **106**. This reduces computing resources and network traffic compared to the user **106** searching through a plurality of listings for individual trading cards, which increases an efficiency of the client device **104**.

[0085] Moreover, the first stage **402** shows the augmented reality digital content **158** as including a price augmentation **412**, only one of which is labeled for illustrative clarity. The price augmentation **412** is generated by the image augmentation service **114** in response to detecting a price tag **414** (only one of which is labeled for illustrative clarity) associated with a detected and identified trading card. By way of example, the price tag **414** is identified by the object detection algorithm **134** and evaluated via the text processing module **146** in order to determine the price shown on the

price tag **414**. Based on the input to the currency selection menu **324** shown in the third stage **306** of FIG. 3, the price augmentation **412** gives the price in USD. It is to be appreciated that the price given on the price tag **414** may be in USD or a different currency.

[0086] Although the first stage **402** includes the price augmentation **412** as including the price on the price tag **414**, in variations, the price augmentation **412** includes additional or alternative price information. By way of example, the price augmentation **412** may additionally or alternatively include an average sale price (e.g., a sold price and/or currently listed price) of the detected and identified trading card in the identified listings, a highest price of the detected and identified trading card in identified listings, and/or a lowest price of the of the detected and identified trading card in the identified listings. As such, the price augmentation **412** may summarize price data in a variety of ways.

[0087] The digital content generation module **120** may configure the augmented reality digital content **158** to remain “as is” relative to the view of the plurality of objects **310**, e.g., based on data received from orientation sensors such as accelerometers, inertial devices, a gyroscope, image processing and feature recognition from the digital images, and so forth. Moreover, the digital content generation module **120** may generate the augmented reality digital content **158** for individual items depicted in the digital images **126** in parallel such that the augmented reality digital content **158** is displayed for multiple detected objects at substantially the same time.

[0088] Still other information may be included in the augmented reality digital content **158**, such as information that does not relate to price. For instance, in at least one implementation, the augmented reality digital content **158** includes a number of current listings for the item ID **218** on the service provider system **102**, a number of listings for the item ID **218** that have been sold on the service provider system **102** within a pre-determined timeframe, an average listing duration before a sale occurs, and so forth.

[0089] As still another example, the augmented reality digital content **158** may be customized for the user **106** based on the user profile **206**. As a non-limiting example, the augmented reality digital content **158** may indicate which of the plurality of objects **310**, if any, are currently owned by the user **106** or are on the user **106**'s wish list based on the user profile **206**. For instance, in an alternative example, the first visual indicator **408** indicates items that are on the user **106**'s wish list, while the second visual indicator **410** indicates items that are not on the user **106**'s wish list. By including simple visual indications for such information in the augmented reality digital content **158**, wish list or collection searching is avoided, which reduces the computing resources uses by the client device **104**.

[0090] The user interface **308** receives input selecting a trading card **416** at the first stage **402**, such as via the finger of the hand **314** tapping the touchscreen at the trading card **416**. In response, at the second stage **404**, the user interface **308** outputs a menu **418**. The menu **418** includes a preview **420** of the trading card **416** as depicted in the catalog **140**, for example, as well as object information **422**. By way of example, the preview **420** and/or the object information **422** is usable by the user **106** to confirm an accurate identification of the trading card **416** by the image augmentation service **114**.

[0091] The menu **418** further includes a first control element **424** and a second control element **426**. In this example, the first control element **424** is selectable to cause generation of price guide information, whereas the second control element **426** is selectable to return to a previous output of the user interface **308**, enabling the user **106** to select a different detected and identified trading card, for example. The second stage **404** shows user input being received at the first control element **424**, such as detected using the touchscreen functionality of the client device **104** at the first control element **424**.

[0092] In response to the user selection of the first control element **424**, a price guide **428** is generated in the user interface **308**, as shown in the third stage **406**. The price guide **428**, for instance, provides a detailed breakdown of the price data used to generate the augmented reality

digital content **158** and is one example of the item guide **160** described with respect to FIG. **1**. In the depicted example, the price guide **428** includes an estimated market value section **430**, a date range selection **432** from among a plurality of selectable date range options **434**, a price chart **436**, and a statistics section **438**. The estimated market value section **430** indicates a predicted sale price of the trading card **416**, as determined based on the listings **224** retrieved for the date range selection **432** (e.g., the desired time period **212**), which is one year in this example. Selection of another of the plurality of selectable date range options **434** may cause at least a portion of the information included in the estimated market value section **430**, the price chart **436**, and/or the statistics section **438** to change.

[0093] The price chart **436** graphically shows a plot of an average sale price of the trading card **416** over time, e.g., over the previous year based on the date range selection **432**. The price chart **436** provides information regarding time-dependent price fluctuations, for example. In the depicted example, the statistics section **438** includes an average sold price for the date range selection **432**, a sold price range for the date range selection **432**, a total number of the trading card **416** sold during the date range selection **432**, and a total number of sellers of the trading card **416** during the date range selection **432**. It is to be appreciated that the statistics section **438** may include more or different information without departing from the scope of the present disclosure.

[0094] The price guide **428** further includes a third control element **440**. The third control element **440** is selectable to close the price guide **428** so that the price guide **428** is no longer overlaid on the digital images **126**, for example. By way of example, in response to receiving user input selecting the third control element **440**, the user interface **308** may return to the second stage **404**. In this way, a large amount of information may be generated from the aspect data **154** and output via various stages of the user interface **308** via the augmented reality digital content **158**.

Example Procedures

[0095] This section describes examples of procedures for a real-time augmented reality item guide. Aspects of the procedures may be implemented in hardware, firmware, software, or a combination thereof. The procedures are shown as a set of blocks that specify operations performed by one or more devices and are not necessarily limited to the orders shown for performing the operations by the respective blocks.

[0096] FIG. **5** is a flow diagram depicting an algorithm as a step-by-step procedure **500** in an example of implementing a real-time augmented reality item guide. In at least one implementation, the procedure **500** is performable by the image augmentation service **114** as a part of the service provider system **102**.

[0097] A digital image depicting at least one item is received by a service platform from a client device (block **502**). By way of example, the digital image may be captured by a camera platform of the client device (e.g., the camera platform **128** of the client device **104** of FIG. **1**) and may be included in a “live view” of a physical environment that includes the at least one item. As such, it is to be appreciated that the digital image may be a sequence of digital images, such as obtained at a frame rate dictated by settings of the camera platform. The client device may communicate the digital image, e.g., the real-time camera feed, to the service platform over a network using a communication module. In at least one variation, however, the digital image is not received from the client device. By way of example, the digital image may be taken from the service platform itself, such as for caching purposes, retrieved from cloud storage, or retrieved from another web service or application.

[0098] One or more additional inputs (e.g., in addition to the digital image) may be received by the service platform from the client device. The additional inputs may include, for example, user profile data, a category of the at least one item, a desired currency for displaying monetary information, and a desired time period for data used in generating augmented reality digital content, as will be elaborated below. In at least one variation, the service platform receives the additional inputs from another source that is separate from the client device (e.g., a web service or

application) and/or generates the additional inputs based on pre-programmed instructions stored in memory.

[0099] A plurality of listings for the at least one item on the service platform is identified via at least one machine learning model based on the digital image (block **504**). By way of example, the at least one machine learning model may be learned via one or more learning approaches, such as supervised learning, unsupervised learning, and/or reinforcement learning, and may be trained to perform object detection, object recognition, and/or classification in order to gain a high-level understanding of the visual information included in the digital image. In at least one implementation, the category is used by a machine learning platform to select which at least one machine learning model to use for processing the digital image.

[0100] In accordance with the described techniques, identifying the plurality of listings for the at least one item on the service platform based on the digital image includes detecting a boundary of the at least one item via an object detection algorithm (block **506**). By way of example, the object detection algorithm (e.g., the object detection algorithm **134** introduced in FIG. **1**) may be trained to locate and identify (e.g., classify) the at least one item depicted in the digital image, such as based on shape. For example, the object detection algorithm may be configured to detect a boundary of the at least one item in the digital image. Separate boundaries may be detected for individual items of the at least one item, for instance.

[0101] Identifying the plurality of listings for the at least one item on the service platform based on the digital image further includes extracting feature vectors from within the detected boundary via a feature extraction algorithm (block **508**). By way of example, the feature extraction algorithm (e.g., the feature extraction algorithm **136** introduced in FIG. **1**) may be trained to identify specific patterns, edges, colors, textures, textual features, and/or other distinctive visual elements of the at least one item depicted in the digital image and detected by the object detection algorithm. The feature vectors extracted by the feature extraction algorithm are numerical representations of the visual elements within the detected boundary for the at least one item. For example, the feature extraction algorithm may output separate feature vectors for individual items of the at least one item based on pixels within respective boundaries.

[0102] Identifying the plurality of listings for the at least one item on the service platform based on the digital image further includes matching the feature vectors to an item listed in the plurality of listings via an item identification algorithm (block **510**). By way of example, the item identification algorithm (e.g., the item identification algorithm **138** introduced in FIG. **1**) matches the feature vectors of the at least one item to a cataloged item having an item identifier (e.g., the item ID **218** of FIG. **2**), and a search module (e.g., the search module **118** introduced in FIG. **1**) identifies the plurality of listings from a listing database using the item identifier. The item identification algorithm, for instance, may compare the feature vectors with feature vectors for cataloged items using a similarity measurement that indicates a degree of similarity. The similarity measurement may use a Euclidean distance, a cosine similarity, and/or a correlation coefficient, for example.

[0103] When a match is identified, e.g., an average similarity measurement is greater than a similarity threshold, the item identifier associated with the matching cataloged item is used by the search module to identify the plurality of listings for the at least one item. The plurality of listings includes one or a combination of active listings for the at least one item on the service platform and inactive listings for the at least one item, e.g., listings that are no longer accessible on the service platform. Each of the plurality of listings is associated with a respective point in time. For instance, the point in time may be a time at which the listing was created, a time at which the listing became inactive, or the like. In examples where the input data includes a desired time period, the identified plurality of listings may be constrained to those having associated time points within the desired time period.

[0104] An aspect is fetched from each of the plurality of listings for the at least one item on the service platform (block **512**). By way of example, the search module includes an aspect retrieval

function (e.g., the aspect retrieval function **156** introduced in FIG. **1**) for fetching aspect data regarding the aspect from each of the plurality of listings in bulk. Non-limiting examples of aspects include price, size, color, model, brand, designer, year of manufacture or release, issue number, and so forth. It is to be appreciated that aspect data for more than one aspect may be retrieved, such as according to preferences stored in the user profile. As such, the aspect data retrieved from the plurality of listings may be associated with the respective points in time in order to generate time-dependent data. The time-dependence, for example, may account for how an aspect, such as the color or price, of the at least one item varies or remains constant with respect to time.

[0105] Augmented reality digital content is generated for the at least one item based on the aspect (block **514**). By way of example, a digital content generation module (e.g., the digital content generation module **120** of FIG. **1**) receives the aspect data and generates the augmented reality digital content (e.g., the augmented reality digital content **158** introduced in FIG. **1**) to provide information regarding the aspect. The augmented reality digital content, for instance, may describe a brand name of the at least one item, a price for which the at least one item is available for sale or purchase (e.g., via the service provider system **102** of FIG. **1**), a relative popularity of the at least one item (e.g., based on a number sold during a specified period of time), and so forth, as determined from the aspect data. Accordingly, the digital content generation module may include functionality for interpreting and/or summarizing the aspect data.

[0106] In at least one implementation, the augmented reality digital content varies as a function of time, e.g., as new data becomes available or data used in generating the digital content becomes out of date. In this context, the augmented reality digital content includes real-time, up-to-date information, and the augmented reality digital content is time-dependent. The augmented reality digital content, for instance, may include a price guide that shows how a monetary value of the object has changed with respect to time. As such, the augmented reality digital content may function as an item guide that provides information regarding the at least one item and/or for purchasing the at least one item.

[0107] The augmented reality digital content is rendered as an overlay on the digital image (block **516**). By way of example, the augmented reality digital content may be broadcast for display at the client device via a user interface and a display device. In at least one variation, the augmented reality digital content is additionally or alternatively rendered at a different device, such as at the service platform itself or at another networked device. The augmented reality digital content may be displayed proximal to the at least one item. In this way, the camera platform enables a user to “look around” the physical environment and display aspect information that describes objects (e.g., the at least one item) included within the physical environment. Moreover, in one or more implementations, the augmented reality digital content includes selectable elements that enable user interaction with the augmented reality digital content. Input received via the selectable elements may cause different or additional information to be displayed, for example.

Example System and Device

[0108] FIG. **6** illustrates an example of a system generally at **600** that includes an example of a computing device **602** that is representative of one or more computing systems and/or devices that may implement the various techniques described herein. This is illustrated through inclusion of the image augmentation service **114**. The computing device **602** may be, for example, a server of a service provider, a device associated with a client (e.g., a client device), an on-chip system, and/or any other suitable computing device or computing system.

[0109] The example computing device **602** as illustrated includes a processing system **604**, one or more computer-readable media **606**, and one or more I/O interfaces **608** that are communicatively coupled, one to another. Although not shown, the computing device **602** may further include a system bus or other data and command transfer system that couples the various components, one to another. A system bus can include any one or combination of different bus structures, such as a memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or

local bus that utilizes any of a variety of bus architectures. A variety of other examples are also contemplated, such as control and data lines.

[0110] The processing system **604** is representative of functionality to perform one or more operations using hardware. Accordingly, the processing system **604** is illustrated as including hardware elements **610** that may be configured as processors, functional blocks, and so forth. This may include implementation in hardware as an application specific integrated circuit or other logic device formed using one or more semiconductors. The hardware elements **610** are not limited by the materials from which they are formed or the processing mechanisms employed therein. For example, processors may be comprised of semiconductor(s) and/or transistors (e.g., electronic integrated circuits (ICs)). In such a context, processor-executable instructions may be electronically executable instructions.

[0111] The computer-readable media **606** is illustrated as including memory/storage **612**. The memory/storage **612** represents memory/storage capacity associated with one or more computer-readable media. The memory/storage **612** may include volatile media (such as random-access memory (RAM)) and/or nonvolatile media (such as read only memory (ROM), Flash memory, optical disks, magnetic disks, and so forth). The memory/storage **612** may include fixed media (e.g., RAM, ROM, a fixed hard drive, and so on) as well as removable media (e.g., Flash memory, a removable hard drive, an optical disc, and so forth). The computer-readable media **606** may be configured in a variety of other ways as further described below.

[0112] Input/output interface(s) **608** are representative of functionality to allow a user to enter commands and information to computing device **602**, and also allow information to be presented to the user and/or other components or devices using various input/output devices. Examples of input devices include a keyboard, a cursor control device (e.g., a mouse), a microphone, a scanner, touch functionality (e.g., capacitive or other sensors that are configured to detect physical touch), a camera (e.g., which may employ visible or non-visible wavelengths such as infrared frequencies to recognize movement as gestures that do not involve touch), and so forth. Examples of output devices include a display device (e.g., a monitor or projector), speakers, a printer, a network card, tactile-response device, and so forth. Thus, the computing device **602** may be configured in a variety of ways as further described below to support user interaction.

[0113] Various techniques may be described herein in the general context of software, hardware elements, or program modules. Generally, such modules include routines, programs, objects, elements, components, data structures, and so forth that perform particular tasks or implement particular abstract data types. The terms “module,” “functionality,” and “component” as used herein generally represent software, firmware, hardware, or a combination thereof. The features of the techniques described herein are platform-independent, meaning that the techniques may be implemented on a variety of commercial computing platforms having a variety of processors.

[0114] An implementation of the described modules and techniques may be stored on or transmitted across some form of computer-readable media. The computer-readable media may include a variety of media that may be accessed by the computing device **602**. By way of example, and not limitation, computer-readable media may include “computer-readable storage media” and “computer-readable signal media.”

[0115] “Computer-readable storage media” may refer to media and/or devices that enable persistent and/or non-transitory storage of information in contrast to mere signal transmission, carrier waves, or signals per se. Thus, computer-readable storage media refers to non-signal bearing media. The computer-readable storage media includes hardware such as volatile and non-volatile, removable and non-removable media and/or storage devices implemented in a method or technology suitable for storage of information thereon, such as computer readable instructions, data structures, program modules, logic elements/circuits, or other data. Examples of computer-readable storage media may include, but are not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, hard disks, magnetic cassettes,

magnetic tape, magnetic disk storage or other magnetic storage devices, or other storage device, tangible media, or article of manufacture suitable to store the desired information and which may be accessed by a computer.

[0116] “Computer-readable signal media” may refer to a signal-bearing medium that is configured to transmit instructions to the hardware of the computing device **602**, such as via a network. Signal media typically may embody computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as carrier waves, data signals, or other transport mechanism. Signal media also include any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared, and other wireless media.

[0117] As previously described, hardware elements **610** and computer-readable media **606** are representative of modules, programmable device logic and/or fixed device logic implemented in a hardware form that may be employed in some embodiments to implement at least some aspects of the techniques described herein, such as to perform one or more instructions. Hardware may include components of an integrated circuit or on-chip system, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and other implementations in silicon or other hardware. In this context, hardware may operate as a processing device that performs program tasks defined by instructions and/or logic embodied by the hardware as well as a hardware utilized to store instructions for execution, e.g., the computer-readable storage media described previously.

[0118] Combinations of the foregoing may also be employed to implement various techniques described herein. Accordingly, software, hardware, or executable modules may be implemented as one or more instructions and/or logic embodied on some form of computer-readable storage media and/or by one or more hardware elements **610**. The computing device **602** may be configured to implement particular instructions and/or functions corresponding to the software and/or hardware modules. Accordingly, implementation of a module that is executable by the computing device **602** as software may be achieved at least partially in hardware, e.g., through use of computer-readable storage media and/or hardware elements **610** of the processing system **604**. The instructions and/or functions may be executable/operable by one or more articles of manufacture (for example, one or more computing devices **602** and/or processing systems **604**) to implement techniques, modules, and examples described herein.

[0119] The techniques described herein may be supported by various configurations of the computing device **602** and are not limited to the specific examples of the techniques described herein. This functionality may also be implemented all or in part through use of a distributed system, such as over a “cloud” **614** via a platform **616** as described below.

[0120] The cloud **614** includes and/or is representative of a platform **616** for resources **618**. The platform **616** abstracts underlying functionality of hardware (e.g., servers) and software resources of the cloud **614**. The resources **618** may include applications and/or data that can be utilized while computer processing is executed on servers that are remote from the computing device **602**. Resources **618** can also include services provided over the Internet and/or through a subscriber network, such as a cellular or Wi-Fi network.

[0121] The platform **616** may abstract resources and functions to connect the computing device **602** with other computing devices. The platform **616** may also serve to abstract scaling of resources to provide a corresponding level of scale to encountered demand for the resources **618** that are implemented via the platform **616**. Accordingly, in an interconnected device embodiment, implementation of functionality described herein may be distributed throughout the system **600**. For example, the functionality may be implemented in part on the computing device **602** as well as via the platform **616** that abstracts the functionality of the cloud **614**.

Conclusion

[0122] Although the systems and techniques have been described in language specific to structural features and/or methodological acts, it is to be understood that the systems and techniques defined in the appended claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed subject matter.

Claims

1. A method for an augmented reality guide, the method comprising: receiving, by a service platform, a digital image depicting at least one item; identifying, via at least one machine learning model of the service platform, a plurality of listings for the at least one item on the service platform based on the digital image, each of the plurality of listings associated with a respective point in time; fetching an aspect from each of the plurality of listings for the at least one item on the service platform; generating time-dependent augmented reality digital content for the at least one item based on the aspect; and rendering the time-dependent augmented reality digital content as an overlay on the digital image.
2. The method of claim 1, wherein the aspect comprises at least one of a color, a size, a brand, a model, or a year of release of the at least one item.
3. The method of claim 1, wherein the aspect comprises price data, and the method further comprises: detecting, via a classifier, a price of the at least one item as depicted in the digital image; and generating the time-dependent augmented reality digital content for the at least one item further based on the detected price.
4. The method of claim 3, wherein the time-dependent augmented reality digital content comprises a first visual indicator in response to the detected price being less than an average of the price data or a second visual indicator in response to the detected price being greater than the average of the price data.
5. The method of claim 3, wherein the time-dependent augmented reality digital content comprises an indication of the price of the at least one item in a different currency than the detected price.
6. The method of claim 3, wherein the time-dependent augmented reality digital content includes a summary of the price data.
7. The method of claim 6, wherein the summary comprises an average price, a highest price, and a lowest price of the plurality of listings.
8. The method of claim 1, wherein identifying, via the at least one machine learning model of the service platform, the plurality of listings for the at least one item on the service platform based on the digital image comprises: detecting a boundary of the at least one item based on the digital image; outputting feature vectors of a portion of the digital image within the detected boundary; and matching the feature vectors to an item listed on the service platform via the plurality of listings.
9. The method of claim 1, wherein the at least one machine learning model of the service platform is trained for a category of items that includes the at least one item, and wherein the category of items is received via a user interface.
10. The method of claim 1, wherein the digital image is received as part of a live camera feed.
11. The method of claim 1, further comprising: receiving a selection of the time-dependent augmented reality digital content; and generating a chart for the aspect with respect to a defined period of time based on aspect data for listings associated with time points within the defined period of time, the defined period of time adjustable based on an input.
12. The method of claim 1, wherein the at least one item comprises a plurality of items, and wherein the time-dependent augmented reality digital content is generated for individual items of the plurality of items in parallel.

13. A computing device comprising: a processing system; and a computer-readable storage medium storing instructions that, responsive to execution by the processing system, causes the processing system to perform operations comprising: displaying a digital image from a live camera feed; determining an item identifier for at least one object depicted in the digital image by processing the digital image using a computer vision module of a service platform; retrieving, based on the item identifier, aspect data from a plurality of listings for the at least one object on the service platform, each of the plurality of listings associated with a respective time point; generating augmented reality digital content for the at least one object based on the aspect data with respect to a specified period of time; and rendering the augmented reality digital content as an overlay on the digital image.

14. The computing device of claim 13, wherein determining the item identifier for the at least one object depicted in the digital image by processing the digital image using the computer vision module of the service platform comprises: detecting a boundary of the at least one object in the digital image; outputting feature vectors of a portion of the digital image within the detected boundary; and matching the feature vectors to an item having the item identifier via a similarity measurement, wherein the plurality of listings includes instances of the item having the item identifier listed for sale on the service platform.

15. The computing device of claim 13, wherein generating the augmented reality digital content for the at least one object based on the aspect data with respect to the specified period of time comprises: receiving, by the service platform, a selection of the specified period of time; constraining the aspect data to a portion generated within the specified period of time based on the respective time point of an associated listing of the plurality of listings; and generating the augmented reality digital content using the constrained aspect data.

16. The computing device of claim 13, wherein the augmented reality digital content indicates at least one of an average sale price, a highest sale price, a lowest sale price, a purchase profitability, a number sold, a color, a size, a brand, a model, or a year of release of the at least one object within the specified period of time.

17. A computer-readable storage medium storing instructions that, responsive to execution by one or more processors, causes the one or more processors to perform operations comprising: receiving, by a service platform, a digital image from a live camera feed; determining an item identifier for an item depicted in the digital image by processing the digital image via at least one machine learning model of the service platform; retrieving a plurality of listings for the item on the service platform based on the item identifier; fetching aspect data from each of the plurality of listings for the item on the service platform, the aspect data associated with points in time corresponding to respective listings of the plurality of listings; generating time-dependent augmented reality digital content for the item based on the aspect data; and rendering the time-dependent augmented reality digital content as an overlay on the digital image.

18. The computer-readable storage medium of claim 17, wherein determining the item identifier for the item depicted in the digital image by processing the digital image via the at least one machine learning model of the service platform comprises: detecting a boundary of the item in the digital image via an object detection algorithm; extracting feature vectors from within the detected boundary via a feature extraction algorithm; and matching the feature vectors to an item having the item identifier via an item identification algorithm and based on a catalog of items, the catalog of items comprising entries for a plurality of items associated with item identifiers.

19. The computer-readable storage medium of claim 17, wherein the operations further comprise: selecting the at least one machine learning model of the service platform based at least in part on a category input.

20. The computer-readable storage medium of claim 17, wherein generating the time-dependent augmented reality digital content for the item based on the aspect data comprises: generating a

chart relating an aspect of the aspect data to time for a defined period of time, the defined period of time adjustable based on an input.
